EMLM_26_Descriptives

Introduction

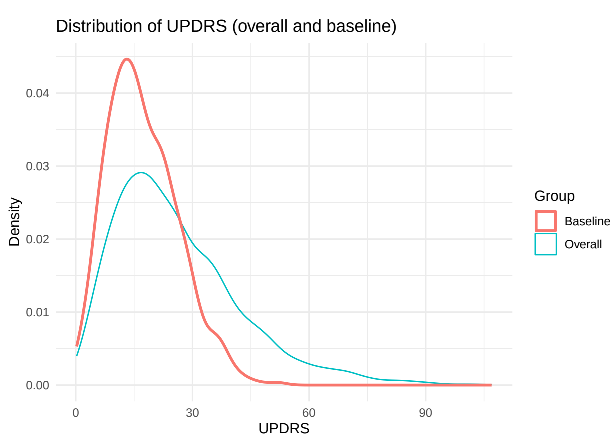
1 Descriptive Statistics

1.1 Summary Statistics for the Sample

The study population consisted of 500 participants with a wide age range (41.1-88 years), although the median age was above 65 years, indicating an overall older cohort. The distribution of sex was balanced, with equal numbers of male and female participants. Smoking status was unevenly distributed across groups, with more than half of the participants classified as never smokers (52.2%), followed by former smokers (28.2%) and a smaller proportion of current smokers (19.6%). The relatively small size of the current smoker group may limit precision when comparing disease trajectories across smoking categories, which will be considered in the subsequent analysis.

Table 1: Descriptive Statistics

Variable	Value
Age (years), mean (SD)	64.64 (13.49)
Age (years), median	65.1
Age (years), min–max	41.1 – 88
Sex: Male, n (%)	250 (50%)
Sex: Female, n (%)	250 (50%)
Smoking status: Never (0), n (%)	261 (52.2%)
Smoking status: Former (1), n (%)	141 (28.2%)
Smoking status: Current (2), n (%)	98 (19.6%)

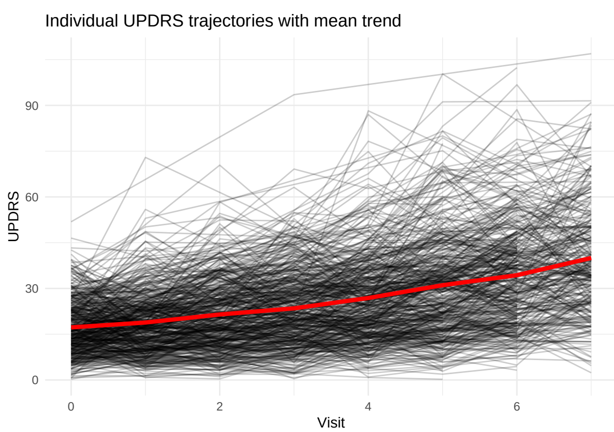


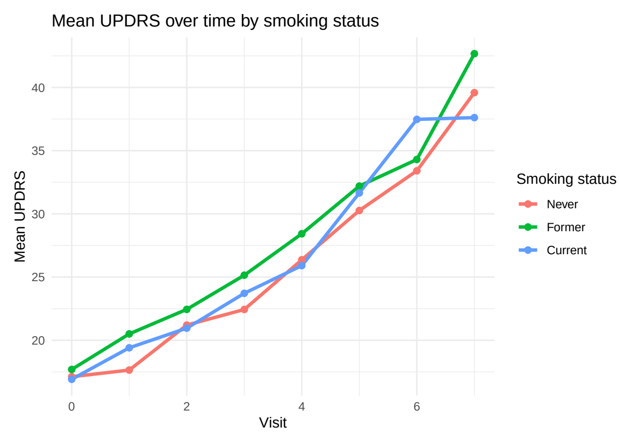
Across all visits, UPDRS scores exhibited a right-skewed distribution, with most observations concentrated in the mild-to-moderate range and a long tail toward higher values. The median score was 22.73, with values ranging from 0.22 to 106.94, indicating a wide range of motor severity across the study period. At baseline, UPDRS scores were lower and less variable, with a median of 16.02 and a narrower range (0.23–51.85), suggesting that most patients entered the study with mild to moderate impairment. The baseline distribution also showed moderate variability at diagnosis and a mild right skew.

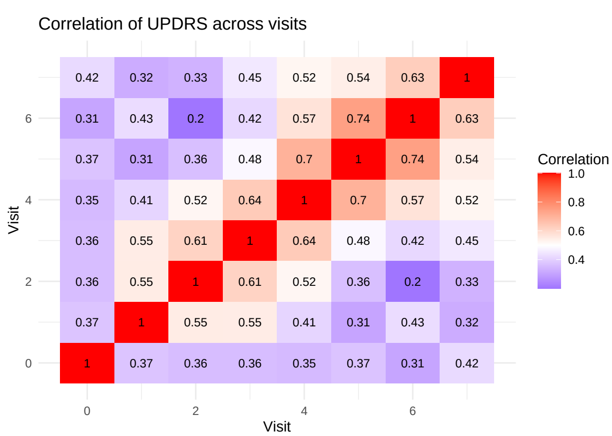
The age distribution was broadly similar across sex groups, with comparable medians (65.2 years for males and 64.1 years for females) and overall variability. No substantial age differences in variability were observed across sex groups(see Appendix A). Similarly, age distributions were comparable across smoking groups, with substantial overlap between categories. Former smokers appeared slightly older on average, while current smokers were marginally younger; however, these differences were small (see Appendix B).

1.2 Progression of Disease Severity (UPDRS)

The relationship between age and UPDRS showed a weak positive association (r = 0.23), with substantial variability in UPDRS scores (SD = 16.28) across all ages. This suggests that age may play a role in disease severity and should be accounted for in the statistical analysis. In contrast, mean UPDRS trajectories over time were highly similar across sex groups, with no clear separation between males and females. Together, these findings do not indicate strong confounding by sex, while age may act as a potential confounder in the relationship between smoking status and disease progression.







The longitudinal distribution of UPDRS scores indicates a clear increase in mean values over time, as illustrated in Figure ... reflecting progressive motor impairment. The increase in mean UPDRS over time appears approximately linear, with only minor deviations at later visits. Substantial variability is observed both within and between patients, as illustrated by the divergent individual trajectories around the overall mean trend. This indicates heterogeneity in disease progression across individuals, as well as within-subject fluctuations over time.

The correlation matrix of UPDRS measurements across visits indicates moderate to strong positive correlations between repeated observations within individuals, confirming the presence of within-subject dependence. Correlations tend to be higher for visits closer in time and lower for more distant visits, although the pattern is not strictly monotonic. This suggests a temporally structured correlation, where measurements within the same individual are related over time. These findings motivate the use of mixed-effects models, which account for such correlation through subject-specific random effects, including both random intercepts and slopes.

Statistical Analysis

```
model_final = lmer(
  UPDRS ~ Visit * smoking_status + Age + sex + (Visit | id),
  data = DF,
  REML = FALSE
)
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, : Model failed to conve
##   See ?lme4::convergence and ?lme4::troubleshooting.
```

```
summary(model_final)
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
## Formula: UPDRS ~ Visit * smoking_status + Age + sex + (Visit | id)
##    Data: DF
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##   21445.8   21516.8  -10710.9   21421.8      2750
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.7749 -0.6250 -0.0965  0.5437  4.5438
##
## Random effects:
##  Groups   Name        Variance Std.Dev. Corr
##  id       (Intercept) 18.778   4.333
##           Visit        3.515   1.875    0.52
##  Residual             94.600   9.726
## Number of obs: 2762, groups:  id, 500
##
## Fixed effects:
##                             Estimate Std. Error t value
## (Intercept)                 -0.92725    1.85155  -0.501
## Visit                        2.99718    0.16123  18.589
## smoking_statusFormer         1.03673    0.87893   1.180
## smoking_statusCurrent        0.67856    0.98877   0.686
## Age                          0.25492    0.02672   9.542
## sexfemale                    0.10564    0.71906   0.147
## Visit:smoking_statusFormer   0.15762    0.27493   0.573
## Visit:smoking_statusCurrent -0.13222    0.31095  -0.425
##
## Correlation of Fixed Effects:
##             (Intr) Visit  smkn_F smkn_C Age    sexfml Vst:_F
## Visit       -0.083
```

```
## smkng_sttsF -0.128  0.168
## smkng_sttsC -0.168  0.149  0.309
## Age         -0.939  0.004 -0.040  0.014
## sexfemale   -0.242 -0.001  0.000  0.044  0.046
## Vst:smkng_F  0.050 -0.586 -0.291 -0.087 -0.004  0.005
## Vst:smkng_C  0.040 -0.519 -0.087 -0.286  0.002 -0.002  0.304
## optimizer (nloptwrap) convergence code: 0 (OK)
## Model failed to converge with max|grad| = 0.002151 (tol = 0.002, component 1)
##   See ?lme4::convergence and ?lme4::troubleshooting.
```

```r
anova(model_final)
```

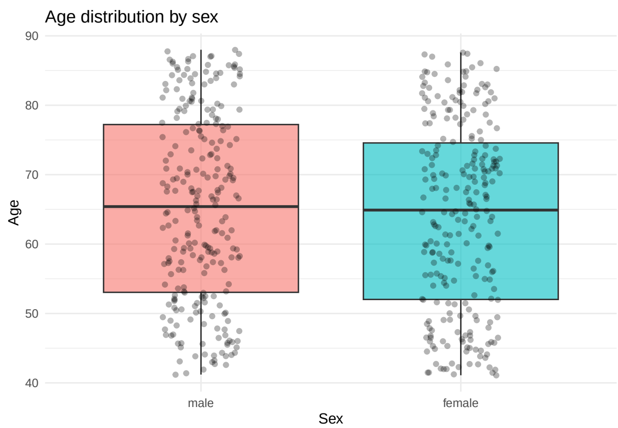
```
## Analysis of Variance Table
##                      npar Sum Sq Mean Sq  F value
## Visit                   1  62463   62463 660.2903
## smoking_status          2    314     157   1.6615
## Age                     1   8628    8628  91.2030
## sex                     1      2       2   0.0201
## Visit:smoking_status    2     69      34   0.3624
```

Assumptions

Appendix

Appendix A: Relationship between Age and Sex This plot describes the distribution of Age by Sex.

```r
ggplot(DF_pat, aes(x = sex, y = Age, fill = sex)) +
  geom_boxplot(alpha = 0.6, outlier.shape = NA) +
  geom_jitter(width = 0.15, alpha = 0.3) +
  labs(
    title = "Age distribution by sex",
    x = "Sex",
    y = "Age"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```



Appendix B: Relationship between Age and Smoking Status This plot describes the distribution of Age by Smoking Status.

```
DF_pat = DF %>%
  distinct(id, .keep_all = TRUE)

ggplot(DF_pat, aes(x = smoking_status, y = Age, fill = smoking_status)) +
  geom_boxplot(alpha = 0.6, outlier.shape = NA) +
  geom_jitter(width = 0.15, alpha = 0.3) +
  labs(
    title = "Age distribution by smoking status",
    x = "Smoking status",
    y = "Age"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

Appendix C: Model specification, evaluation, and selection The model building strategy, started with `model_a`, and proceeded with `model_b`, and finally `model_c`. `model_a` tests a fuller fixed-effects structure by including both time-by-smoking and time-by-sex interactions, `model_b` assesses whether the time-by-sex interaction can be removed while retaining random slopes, and `model_c` evaluates whether the random slope for time is necessary at all. For valid model comparison, we fitted all models with maximum likelihood and `lmer()` from the `lme4` package, after which the final selected model can be refitted with REML.

```
# Fuller model: random intercept + random slope
model_a <- lmer(
  UPDRS ~ Visit * smoking_status + Visit * sex + Age + (Visit | id),
  data = DF,
  REML = FALSE
)

# Without Visit:sex interaction, still random intercept + slope
model_b <- lmer(
  UPDRS ~ Visit * smoking_status + Age + sex + (Visit | id),
  data = DF,
  REML = FALSE
)
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, : Model failed to conve
```

```
##   See ?lme4::convergence and ?lme4::troubleshooting.
```

```r
# Same fixed effects as model_b, but random intercept only
model_c <- lmer(
  UPDRS ~ Visit * smoking_status + Age + sex + (1 | id),
  data = DF,
  REML = FALSE
)

anova(model_a, model_b)
```

```
## Data: DF
## Models:
## model_b: UPDRS ~ Visit * smoking_status + Age + sex + (Visit | id)
## model_a: UPDRS ~ Visit * smoking_status + Visit * sex + Age + (Visit | id)
##         npar   AIC   BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## model_b   12 21446 21517 -10711     21422
## model_a   13 21448 21525 -10711     21422 0.1179  1     0.7313
```

```r
AIC(model_a, model_b)
```

```
##         df      AIC
## model_a 13 21447.65
## model_b 12 21445.76
```

```r
BIC(model_a, model_b)
```

```
##         df      BIC
## model_a 13 21524.65
## model_b 12 21516.85
```

```r
anova(model_c, model_b)
```

```
## Data: DF
## Models:
## model_c: UPDRS ~ Visit * smoking_status + Age + sex + (1 | id)
## model_b: UPDRS ~ Visit * smoking_status + Age + sex + (Visit | id)
##         npar   AIC   BIC logLik -2*log(L) Chisq Df Pr(>Chisq)
## model_c   10 21788 21847 -10884     21768
## model_b   12 21446 21517 -10711     21422 345.7  2  < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
AIC(model_c, model_b)
```

```
##         df      AIC
## model_c 10 21787.46
## model_b 12 21445.76
```

```r
BIC(model_c, model_b)
```

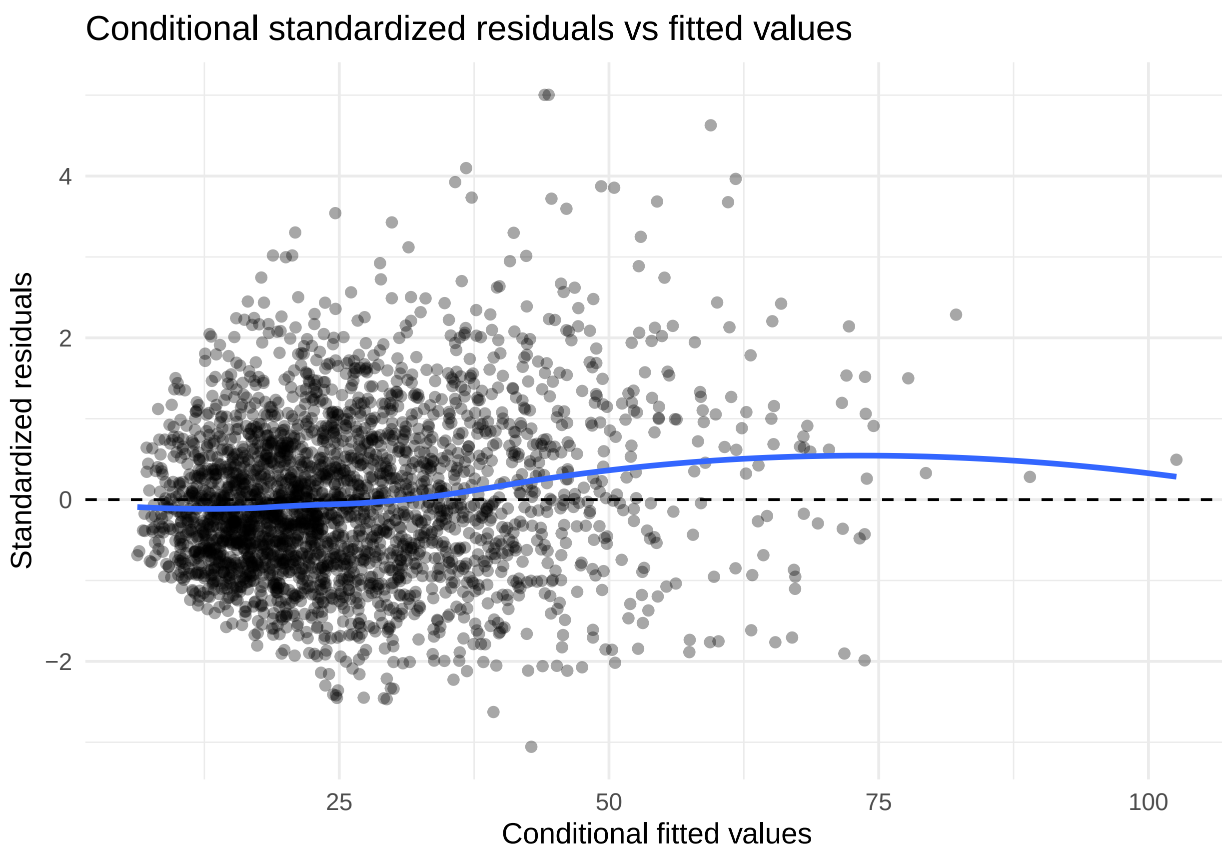
```
##         df      BIC
## model_c 10 21846.70
## model_b 12 21516.85
```

```r
# Setup for diagnostics
diag_df <- DF %>%
  mutate(
    fitted_cond = fitted(model_final),              # conditional fitted values
    resid_cond = resid(model_final),                # conditional/raw residuals
    resid_cond_std = scale(resid_cond)[, 1]     # standardized version
  )

# Conditional Residuals Vs Fitted Values
ggplot(diag_df, aes(x = fitted_cond, y = resid_cond_std)) +
  geom_point(alpha = 0.35) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_smooth(method = "loess", se = FALSE) +
  labs(
    title = "Conditional standardized residuals vs fitted values",
    x = "Conditional fitted values",
    y = "Standardized residuals"
  ) +
  theme_minimal()
```

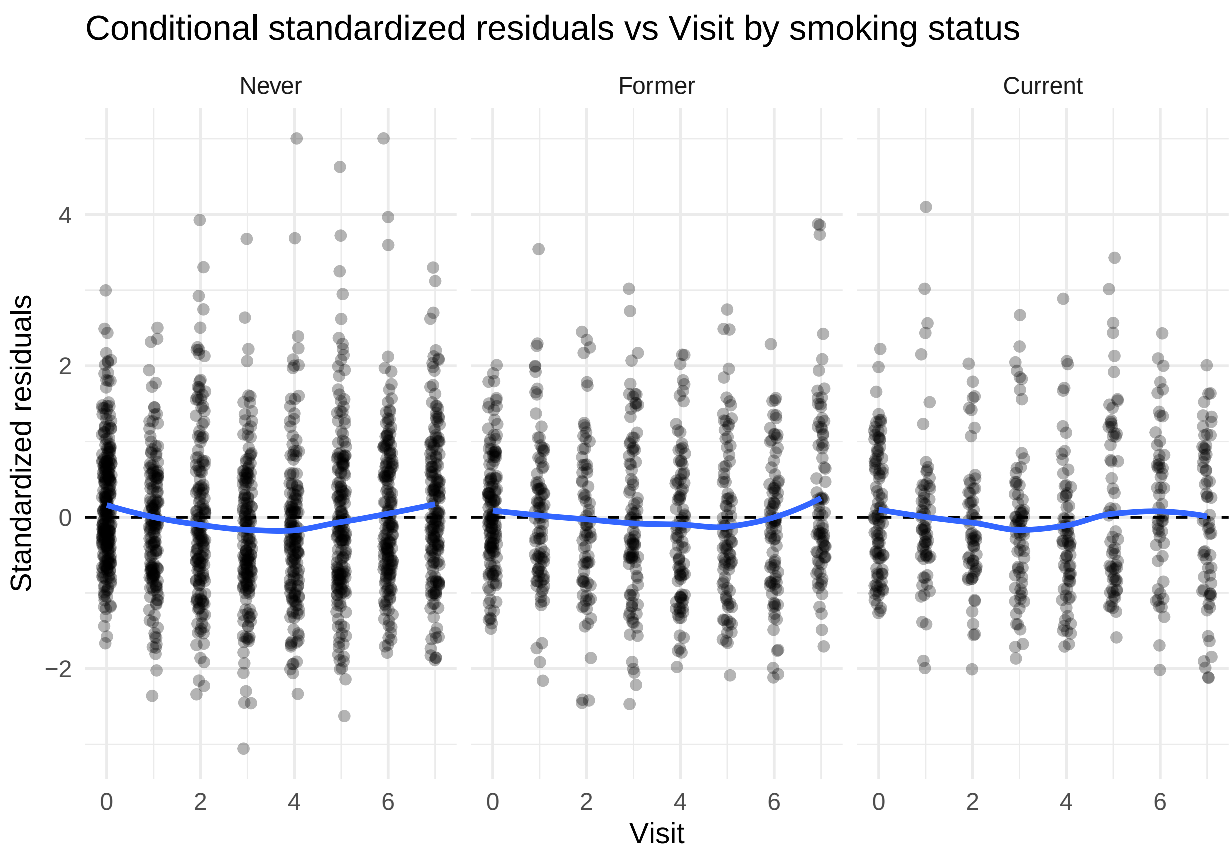
Appendix D: Model Diagnostics

```
## `geom_smooth()` using formula = 'y ~ x'
```

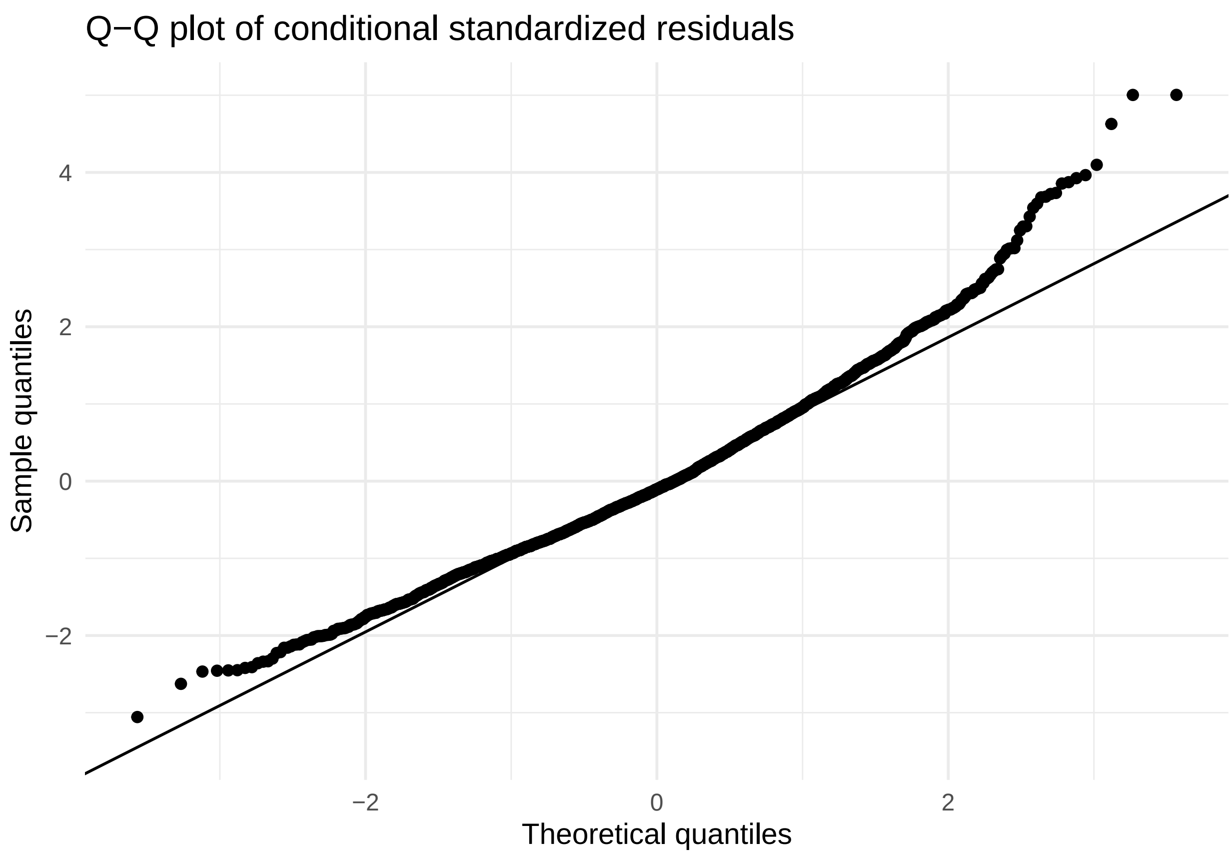



```r
# Conditional Residuals Vs Time by Smoking Group
ggplot(diag_df, aes(x = Visit, y = resid_cond_std)) +
  geom_point(alpha = 0.3, position = position_jitter(width = 0.1, height = 0)) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_smooth(method = "loess", se = FALSE) +
  facet_wrap(~ smoking_status) +
  labs(
    title = "Conditional standardized residuals vs Visit by smoking status",
    x = "Visit",
    y = "Standardized residuals"
  ) +
  theme_minimal()
```

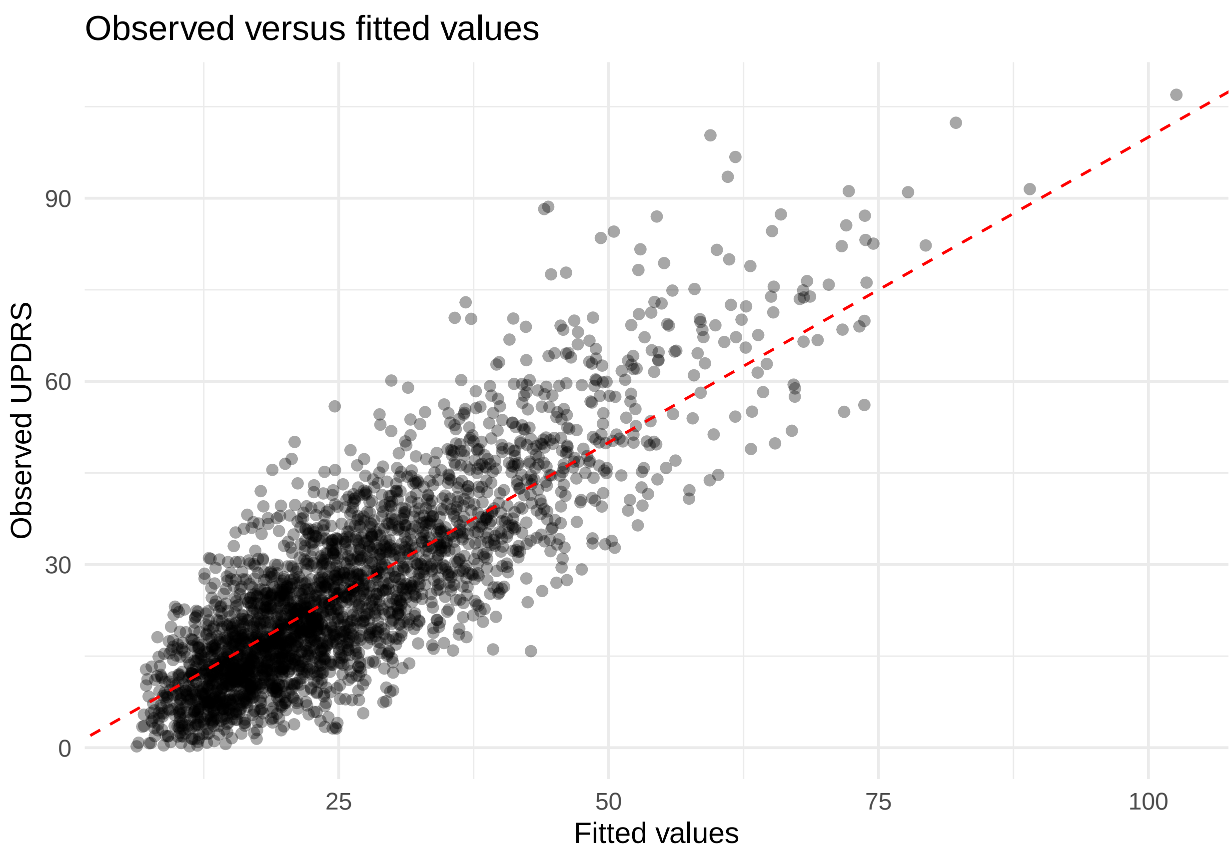
```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
# Q-Q Plot of Conditional Residuals
ggplot(diag_df, aes(sample = resid_cond_std)) +
  stat_qq() +
  stat_qq_line() +
  labs(
    title = "Q-Q plot of conditional standardized residuals",
    x = "Theoretical quantiles",
    y = "Sample quantiles"
  ) +
  theme_minimal()
```

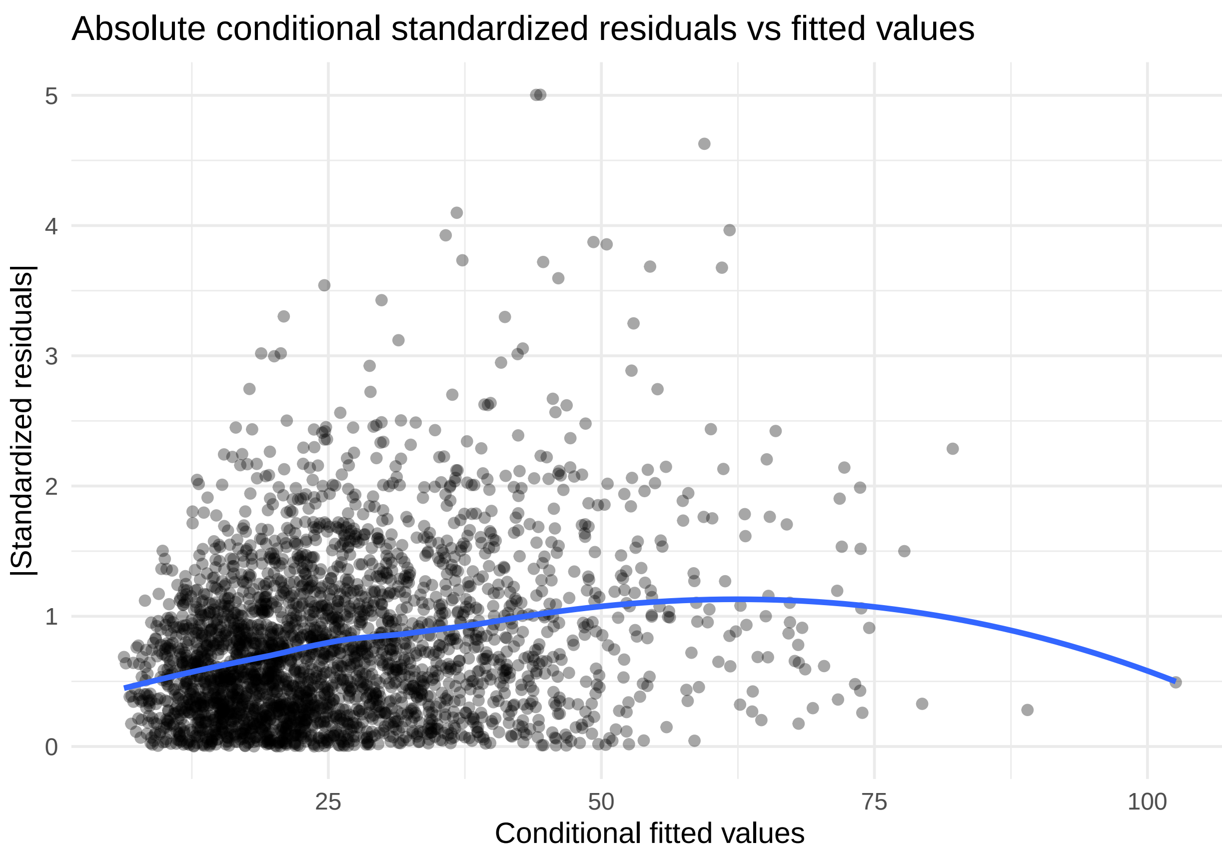



```r
# Observed Vs Fitted Values
ggplot(diag_df, aes(x = fitted_cond, y = UPDRS)) +
  geom_point(alpha = 0.35) +
  geom_abline(intercept = 0, slope = 1, linetype = "dashed", color = "red") +
  labs(
    title = "Observed versus fitted values",
    x = "Fitted values",
    y = "Observed UPDRS"
  ) +
  theme_minimal()
```



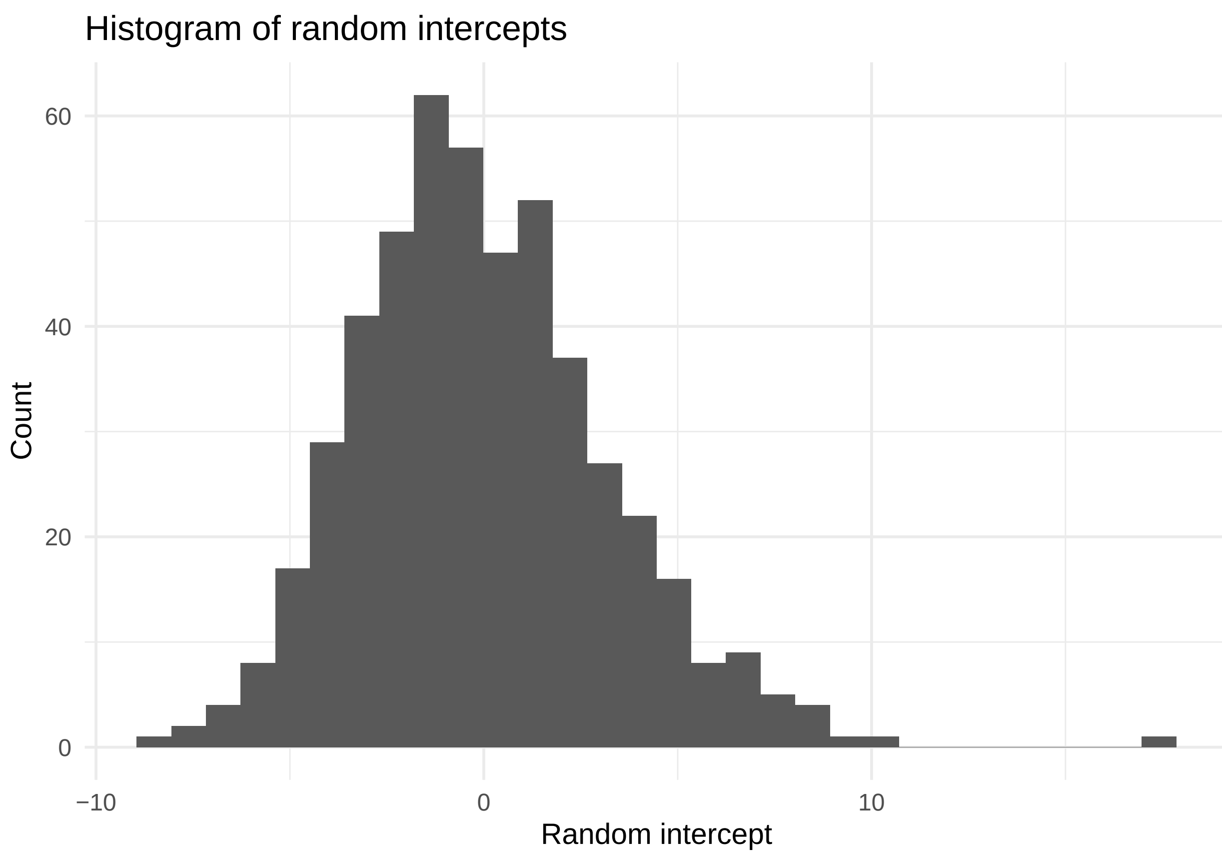

```r
# Heteroscedasticity
ggplot(diag_df, aes(x = fitted_cond, y = abs(resid_cond_std))) +
  geom_point(alpha = 0.35) +
  geom_smooth(method = "loess", se = FALSE) +
  labs(
    title = "Absolute conditional standardized residuals vs fitted values",
    x = "Conditional fitted values",
    y = "|Standardized residuals|"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

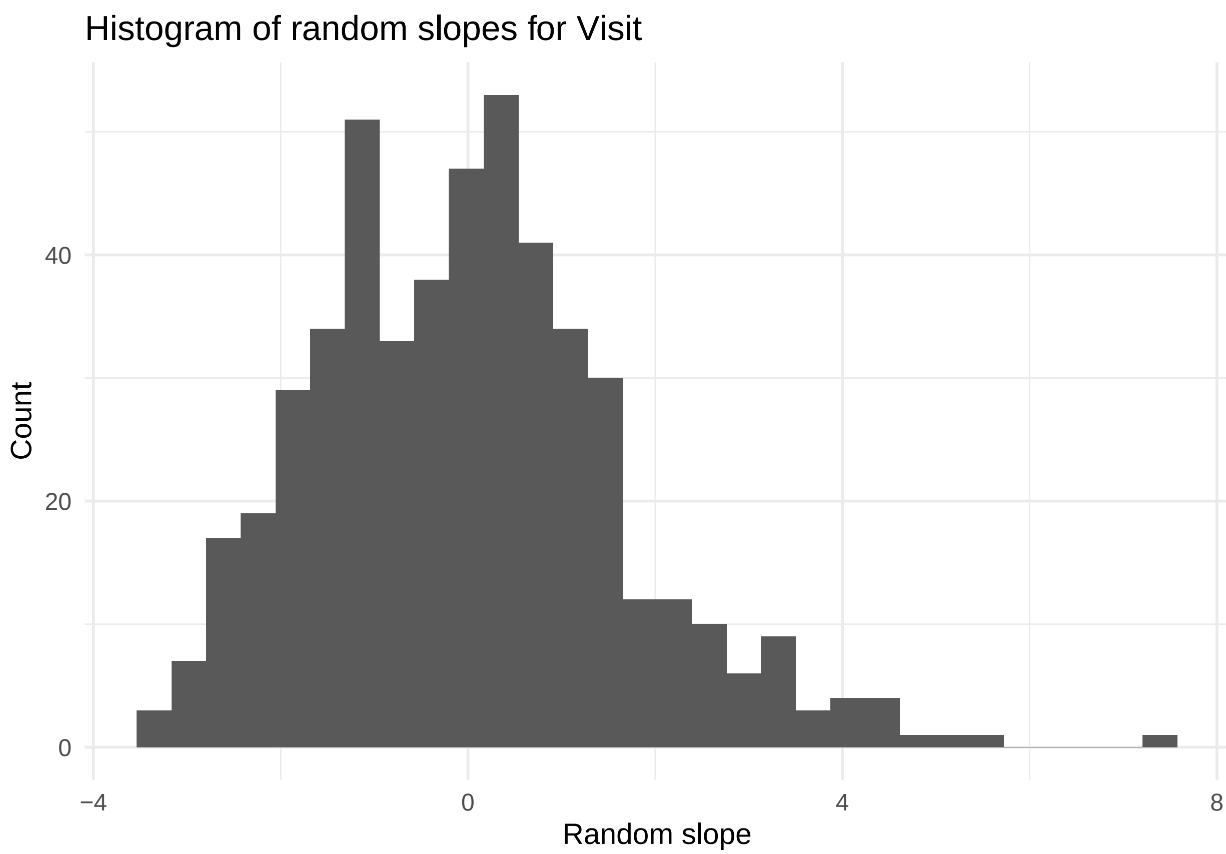



```
# Extract random Effects
re <- ranef(model_final)$id
re$id <- rownames(re)
colnames(re)[1:2] <- c("b_intercept", "b_slope")

# Histogram of random Intercepts
ggplot(re, aes(x = b_intercept)) +
  geom_histogram(bins = 30) +
  labs(
    title = "Histogram of random intercepts",
    x = "Random intercept",
    y = "Count"
  ) +
  theme_minimal()
```




```r
# Histogram of random Slopes
ggplot(re, aes(x = b_slope)) +
  geom_histogram(bins = 30) +
  labs(
    title = "Histogram of random slopes for Visit",
    x = "Random slope",
    y = "Count"
  ) +
  theme_minimal()
```



Several diagnostic plots were examined to assess the assumptions of the linear mixed-effects model.

The residuals vs fitted values plot (to assess linearity and mean structure) shows a slight curvature in the LOESS line, suggesting mild non-linearity. While the deviation is not severe, it indicates that the linear effect of some predictors (possibly Visit or Age) may not fully capture the relationship with UPDRS. **maybe here we have to use a spline**

The residuals vs Visit by smoking status plot (used to check time trends and interaction adequacy) shows generally centered residuals around zero across visits and smoking groups. However, small systematic deviations over time are visible, particularly a slight upward trend around mid visits, suggesting that the time effect may not be perfectly specified (e.g., potential non-linearity in Visit).

The Q–Q plot of residuals (used to assess normality) shows clear deviation in the upper tail, with heavier-than-expected positive residuals. This indicates some departure from normality, likely due to a few large positive residuals (potential outliers or skewness in UPDRS).

The observed vs fitted values plot (used to assess overall model fit) shows that the model captures the general trend well, with points roughly aligned along the identity line. However, variability increases for larger fitted values, suggesting reduced predictive accuracy at higher UPDRS levels. **see below**

The absolute residuals vs fitted values plot (used to assess homoscedasticity) indicates increasing spread in residuals as fitted values increase, followed by a slight decrease at the highest values. This suggests mild heteroscedasticity, with larger variability in mid-to-high fitted ranges. **we probably have to change the variance structure, for instance adding weights to allow variance to increase with fitted values, or (and it's easier, also use a spline, like ns()**

The histogram of random intercepts (used to assess normality of random effects) appears approximately symmetric and bell-shaped, supporting the normality assumption for random intercepts, although a slight right tail is present. **this is fine**

Similarly, the histogram of random slopes for Visit shows an approximately normal distribution with some right skew and a few extreme values, suggesting that while the normality assumption is broadly reasonable, there may be some influential subjects with atypical progression rates. **here we have to check outliers and influential points**

Overall, we see an upward trend that was already visible in the initial diagnostics, where we determined that there was a slight non-linearity. Additionally, we have to deal with non-normality and possible influential (or maybe not, if we can argument or transform - sometimes medical data can benefit of a log transformation, I've done it in the past) outliers.